MILITARY SPECIFICATION

ARMING-FIRING DEVICE, ROCKET MOTOR, MARK 28 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of inertial arming-firing device.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-575	Braid, Wire (Copper, Tin-Coated or Silver-Coated, Tubular, or Flat)

MILITARY

MIL-P-116	Preservation-Packing, Methods of
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-P-19834	Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for
MIL-S-81271	Squib, One Ampere, No Fire

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Documentation Division (501), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Requirements
MS 29513	Packing, Preformed, Hydrocarbon Fuel Resistant, "O" Ring

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (Code Ident 10001)

2177161	Blast Tube Assembly
2177165	Igniter Assembly, Rocket Motor Mark 272 Mod 0 (Rocket Motor Mark 30 Mod 2)
2177166	Arming-Firing Device, Rocket Motor, Mark 28 Mod 0, Source Control Drawing

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Preproduction sample. When specified in the contract (see 6.2), the contractor shall furnish sample units for preproduction inspection and approval (see 4.4 and 6.3).

3.2 Squibs. The electric squibs shall conform to MIL-S-81271. The two squibs shall be electrically connected in parallel.

3.3 Cable. The electrical lead shall be a two-conductor flat shielded cable containing AWG 22 conductor per MIL-W-16878, Type E, with silver-plated copper shielding in accordance with QQ-B-575, overwrapped with a primary cover of skived TFE teflon tape and a secondary cover of FEP teflon. The cable cross section shall have an envelope of no greater than 0.210 inch width and 0.110 inch thickness.

3.4 Design.

3.4.1 Safe and arming positions. The arming-firing device shall have provisions for the following three positions:

a. **Safe.** Electrically safe, mechanically safe
b. **Test armed.** Electrically armed, mechanically safe
c. **Armed.** Electrically armed, mechanically armed

3.4.1.1 Safe position. Under normal circumstances the unit shall be in the electrically safe and mechanically safe position.

3.4.1.2 Test armed. The electrical circuit shall be designed to allow manual closing for resistance checking in the mechanically safe condition. When not held in the test armed condition, the unit shall automatically revert to the electrically safe position.

3.4.1.3 Armed position. The unit shall be in the armed position (electrically armed and mechanically armed) after being subjected to the "g" loadings of 3.5.1.

3.4.2 Mechanical train position. The mechanical train shall be in an out-of-line position when the electrical circuit is in the safe condition or in the test-armed condition. The mechanical train shall be in an in-line, or firing position when the electrical circuit is in the armed condition.

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3.5 Performance.

3.5.1 Arming time. The unit shall arm in not more than 2.50 seconds while being subjected to an acceleration of 16.0 ± 0.5 gravitational units (g) along the longitudinal axis in the direction of forward motion imposed concurrently with a lateral acceleration of 5.0 ± 0.5g.

3.5.2 Non-arming. The unit shall not arm when subjected to an acceleration of 6.5 ± 0.5g along the longitudinal axis in the direction of forward motion for not less than 20 seconds.

3.5.3 Disarming. The armed unit shall not disarm when subjected to a deceleration of 10.0 ± 0.5g along the longitudinal axis in the direction of forward motion for not less than 20 seconds.

3.5.4 Fail-safe diaphragm. There shall be no gas leak through or around the "fail-safe" diaphragm if one or both squibs should be fired while in the mechanically safe condition.

3.5.5 Output. If one or both squibs are fired in the armed position, the output shall be sufficient to initiate the igniter when assembled in accordance with Drawing 2177165.

3.6 Environmental requirements

3.6.1 Temperature cycling. The unit shall perform as specified herein after being subjected to three temperature cycles. Each cycle shall be from ambient temperature to -10° F to ambient temperature, to 135° F, and back to ambient temperature. The unit shall be stabilized at a given temperature for a minimum of one hour prior to proceeding to the next temperature event.

3.6.2 Immersion (leakage). The unit shall show no evidence of leaks when tested in accordance with 4.6.2.5.

3.6.3 Fungus resistance. The unit shall perform as specified herein after exposure to the fungus resistance test in accordance with MIL-STD-810.

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3.6.4 Sand and dust. The unit shall perform as specified herein after exposure to sand and dust in accordance with MIL-STD-810.

3.6.5 Salt spray. The unit shall perform as specified herein after exposure to salt spray in accordance with MIL-STD-202, test condition B, method 101D.

3.6.6 Vibration. The unit shall perform as specified herein after vibration in accordance with 4.6.1.5.

3.6.7 Six-foot drop. The unit shall be safe to handle and dispose of after being tested in accordance with 4.6.1.6.

3.6.8 Operational shock. The unit shall perform as specified herein after being subjected to a shock in accordance with 4.6.1.7.

3.7 Dimensions. All dimensions shall be in accordance with Drawing 2177166.

3.8 Resistance. Resistance shall be as specified on Drawing 2177166.

3.9 Workmanship. The units shall be uniform in quality and shall be clean and free from burrs, sharp edges, and foreign material that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwsie specified in the contract, the contractor may utilize his own facilities or any other facilities available for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. **Temperature**: Room ambient 18° to 35°C (65° to 95° F)
b. **Altitude**: Normal ground
c. **Vibration**: No requirement
d. **Humidity**: Room ambient to 95% relative (max)

4.4 Preproduction inspection. Unless otherwise specified in the contract (see 6.2), preproduction inspection is required and shall be performed at an activity designated by the procuring activity. Preproduction inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. Failure of a sample to comply with any requirement specified herein shall be cause for rejection of the preproduction sample. Further production of arming-firing devices by the contractor prior to approval of the preproduction sample shall be at the contractor's risk.

4.4.1 Sample size. Sixteen arming-firing devices shall be subjected to preproduction inspection.

4.4.2 Inspection routine. The preproduction samples shall undergo the tests specified in TABLE I. Unless otherwise specified, all tests shall be made with the device installed in an Igniter Assembly in accordance with Drawing 2177165.

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TABLE I. Preproduction inspection schedule.

No. of units	Environmental test requirements	Post environmental test requirements	Enviro. and post enviro. test temp.	Notes
4	4.6.1.1, 4.6.1.5, 4.6.1.7	4.6.2.1.2, 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.1.9	-10° F	(1)
2	4.6.1.1, 4.6.1.5, 4.6.1.7	4.6.2.1.2, 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.1.8	-10° F	(1)
4	4.6.1.1, 4.6.1.5, 4.6.1.7	4.6.2.1.2, 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.1.9	135° F	(1) (2)
2	4.6.1.1, 4.6.1.5, 4.6.1.7	4.6.2.1.2, 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.1.8	135° F	(1)
2	4.6.1.1, 4.6.1.5, 4.6.1.7, 4.6.1.2, 4.6.1.3, 4.6.1.4	4.6.2.1.2, 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.1.9	77° F	(1) (2)
1	4.6.1.1, 4.6.1.5, 4.6.1.7, 4.6.1.2, 4.6.1.3, 4.6.1.4	4.6.2.1.2, 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.1.8	77° F	(1)
1	4.6.1.6	Visual examination	135° F and -10°F	

Notes

(1) In order to factorially evaluate any adverse effects resulting from a given environmental test, functional requirements as specified in 4.6.2.1.2, 4.6.2.2, 4.6.2.3 and 4.6.2.4 shall be made following each environmental test as specified in 4.6.1.1, 4.6.1.5 and 4.6.1.7.

(2) Do not de-activate one squib for output test as specified in 4.6.1.9.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.6.2.

4.5.1 Sampling. Except as otherwise specified, 100 percent inspection shall be used for the tests of 4.6.2.1 through 4.6.2.5. The sample size for the destructive tests (4.6.2.6) shall be in accordance with MIL-STD-105, level S-1, normal inspection, single sampling with an AQL of 6.5 percent.

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4.5.2 Inspection lot. An inspection lot shall consist of all units presented for acceptance at one time, produced by a single manufacturer in a single manufacturing sequence under homogeneous conditions of manufacture.

4.6 Methods of inspection.

4.6.1 Environmental procedures.

4.6.1.1 Temperature cycling. The test procedure for temperature cycling shall be in accordance with 3.6.1.

4.6.1.2 Fungus resistance. The test procedure for fungus resistance shall be in accordance with MIL-STD-810, method 508.1.

4.6.1.3 Sand and dust. The test procedure for resistance to sand and dust shall be in accordance with MIL-STD-810, method 510.1.

4.6.1.4 Salt spray. The test procedure shall be in accordance with MIL-STD-202, test method 101A, test conditions B, 5 percent salt solution.

4.6.1.5 Vibration. The unit shall be mounted horizontally on a vibration table as shown in FIGURE 1, and vibrated to the following schedule:

Event	Frequency (Simple harmonic motion)	Acceleration (g)	Time (Minutes)
1	Schedule per MIL-STD-167-1, type I paragraph 3.1		
2	7	3.2	10
3	7	3.2	50

The assembly shall then be vibrated for 2.0 ± 0.5 minutes using an exponential frequency sweep from 20 to 600 hertz (Hz). Peak force shall be equivalent to a 3.0 ± 0.5 g acceleration imposed about a 1.0 ± 0.5 g steady state acceleration. The critical frequency shall be noted and the assembly vibrated at this critical frequency for an additional 6 to 7 seconds, maintaining the same peak force utilized during the frequency sweep.

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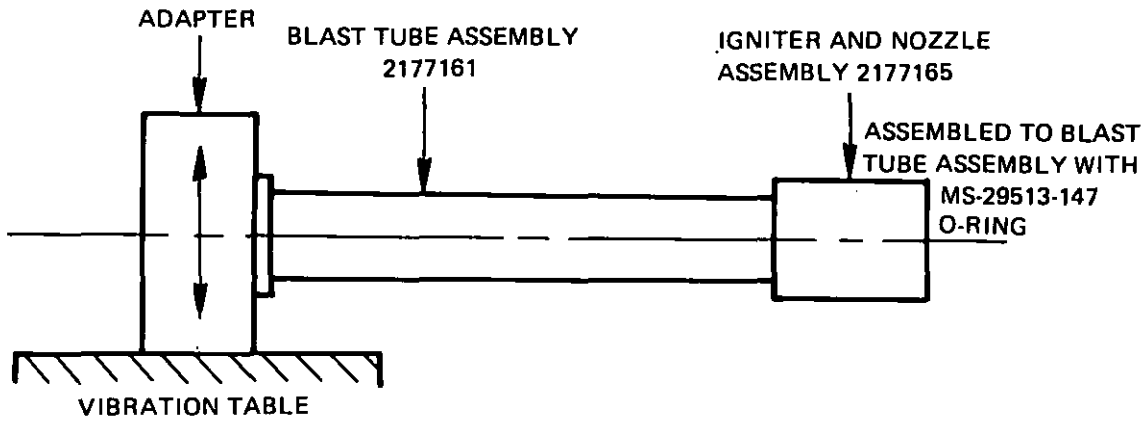


FIGURE 1. Vibration set-up.

4.6.1.6 Six-foot drop. The unit shall be dropped six feet onto a 1/2-inch thick steel anvil resting on at least 6 inches of concrete. The unit shall be guided during the drop so that impact occurs with the base of the arming-firing device facing the anvil. The cable shall be tied so it will not interfere with the impact.

4.6.1.7 Operational shock. A Hyge Acceleration Tester, Consolidated Electrodynamics Corporation, Pasadena, California, or equivalent, shall be used to obtain a shock value of 60.0 ± 9.0 g, having a 0.005 ± 0.003 second rise time and a 0.010 second minimum dwell time, applied along the longitudinal axis in the direction of the forward motion.

4.6.1.8 Fail-safe diaphragm. The unit shall be locked in the test armed position so that ignition current can flow from the connector through the squib circuit while the unit is in the safe position. The squibs shall be initiated by a 6.0 ± 0.1 ampere current. After firing, the unit shall be examined to determine compliance with the requirements of 3.5.4. There shall be no evidence of gas leakage through or around the "fail-safe" diaphragm.

4.6.1.9 Output. With the unit in the armed position and one squib disconnected from the circuit, the other squib shall be initiated by a 3.0 ± 0.5 ampere current supplied through the connector pins. After firing, the unit shall be examined to determine compliance with the requirements of 3.5.5.

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4.6.1.10 Immersion (leakage). The immersion (leakage) shall be determined in accordance with 4.6.2.5.

4.6.2 *Post environmental procedures.*

4.6.2.1 Final inspection.

4.6.2.1.1 Visual examination. Visually examine all units to determine compliance with the requirements of 3.7, 3.9 and Section 5.

4.6.2.1.2 Resistance examination. Each unit shall be checked for resistance to determine compliance with 3.8.

4.6.2.2 Arming time. Each unit shall be spun in a horizontal centrifuge. A holding fixture shall orient the unit with its longitudinal axis in a horizontal plane and at an angle of arctan 5/16 with a radius of the centrifuge and rotated about the center of mass of the G weight in the safe position. The unit shall be in the safe condition at the start of the test and shall be oriented so that radial acceleration will tend to arm it. The distance from the center of rotation of the centrifuge to center of mass of the G weight shall not be less than 12 inches when the unit is in a safe position. The G weight shall be immobilized until the specified value of acceleration is reached at which time the weight shall be released. Coincident with the release of the G weight, a timer shall be started to time total arming. The timer shall be accurate within ±0.02 second. The centrifuge shall produce a value of acceleration accurate to ±0.5g of the absolute value of the acceleration being measured.

4.6.2.3 Disarming. The test procedure shall be as specified in 4.6.2.2, except that the unit shall be in the armed condition at the start of the test and shall be oriented such that the longitudinal axis of the unit is coincident with a radius of the centrifuge and so that radial acceleration will tend to disarm the unit.

4.6.2.4 Non-arming. The test procedure shall be as specified in 4.6.2.2, except that the orientation of the unit shall be such that the longitudinal axis of the unit is coincident with a radius of the centrifuge and that radial acceleration will tend to arm the unit.

4.6.2.5 Immersion (leakage). Place unit in a chamber and then pressurize chamber to 15 ± 1 psig. Maintain this pressure for a minimum of 5 minutes. Fill the chamber with water until the unit is totally submerged. Relieve the pressure in the chamber to ambient. Observe the unit for a minimum of one minute. Any evidence of leakage in the form of a

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continuous stream of bubbles or any bubble (s) which steadily increase in size shall be cause for rejection of the unit. Bubbles which appear on the surface of the unit but which do not increase in size shall not be cause for rejection of the unit.

4.6.2.6 Destructive tests. One half the sample shall be subjected to the functioning test and the other half to the fail-safe test. If the sample is not an even number the odd unit shall be used in the functioning test.

4.6.2.6.1 Functioning test. The device shall be assembled as an Igniter Assembly in accordance with Drawing 2177165 less the Nozzle Assembly. The device will then be armed and fired with a 6.0 ± 0.5 ampere current (see 3.5.5).

4.6.2.6.2 Fail-safe test. The out-of-line safety test of 4.6.1.8 shall be run on the remaining units to determine compliance with 3.5.4. The device shall be assembled as an Igniter Assembly in accordance with Drawing 2177165 less the Nozzle Assembly (see 3.5.4).

5. PACKAGING

5.1 Preservation, packaging and packing. Unless otherwise specified by the procuring activity (see 6.2), the preservation, packaging and packing shall be in accordance with 49 CFR 100-199 and MIL-P-116, method III.

5.2 Marking. In addition to any special marking required by the contract (see 6.2), unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and 49 CFR 100-199. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name
b. Material trade name
c. Gross and net weights
d. Lot number and date of manufacture
e. Number and revision letter of this specification
f. Hazard notice
g. Unusual storage requirements

5.2.1 Marking plates. Marking plates shall be in accordance with MIL-P-19834 and shall include contractor's name, part number, drawing and revision letter and serial number.

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6. NOTES

6.1 Intended use. The unit is intended for use in the Mark 272 Mod 0 igniter for the Mark 30 rocket motor.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Place of inspection
c. Maximum lot size, if applicable (see 4.5.2)
d. Place of delivery
e. Quantity required
f. Whether a preproduction sample is required (see 3.1)
g. Request for test data
h. Inspection conditions, if other than as specified (see 4.3)
i. Location of preproduction inspection (see 4.4)
j. Packing, packaging, and marking requirements, if other than as specified (see 5.1 and 5.2).

6.3 Preproduction sample. When a preproduction sample is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The preproduction sample should consist of 16 units. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the preproduction sample.

6.4 Supersession information. This specification includes the requirements of and supersedes Naval Sea Systems Command WS 6436, dated 12 October 1965 with Amendment 2, dated 29 April 1966.

6.5 Preproduction sample waiver. Preproduction samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional preproduction sample inspection. When the preproduction sample is waived, the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved preproduction sample shall determine the minimum requirements of the current contract or order.

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6.6 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the Defense Acquisition Regulation (DAR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

Custodian:

Navy – OS

Preparing activity:

Navy – OS
Project No. 1340-N570

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MIL-A-82690(OS) Arming-Firing Device, Rocket Motor, Mark 28 Mod 0

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